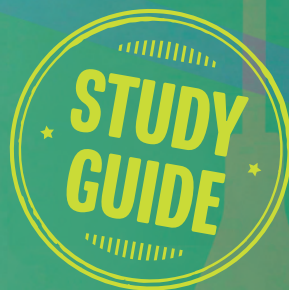


SECOND EDITION

PERIODONTAL REVIEW Q&A

Deborah A. Termeie, DDS



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Periodontal Review Q&A

Second Edition

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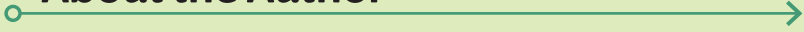
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About the Author



Deborah A. Termeie, DDS, is a clinical instructor in the Department of Periodontics at the University of California, Los Angeles. She is a diplomate of the American Board of Periodontology (ABP), and it was her experience preparing for the ABP qualifying exams that inspired her to write this book. Dr Termeie is also an editor and author of *Avoiding and Treating Dental Complications: Best Practices in Dentistry* (John Wiley & Sons, 2016) and has published on the topics of evidence-based dentistry and implantology. She is the recipient of several awards, including the Excellence in Implantology Research award from the California Society of Periodontics, and has been invited to participate in the American Academy of Periodontics Leadership, Engagement, Action, and Development (LEAD) Program. She maintains a private practice in Beverly Hills, California.

Contents

Preface vi

1	Evidence-Based Dentistry	1
2	Periodontal Anatomy	6
3	Furcations	17
4	Epidemiology and Etiology	31
5	Pharmacology	53
6	Diagnosis	81
7	Prognosis	113
8	Occlusion	125
9	Nonsurgical Therapy	139
10	Surgical Therapy	163
11	Mucogingival Therapy	186
12	Regeneration	221
13	Implants	252
14	Inflammation	299
15	Oral Medicine	311
16	Oral Pathology	326
17	Lasers	347
18	Medical Emergencies	354
19	Treatment Planning	362

Appendix 408

Index 417

Preface

The first edition of *Periodontal Review Q&A* was very well received, and I am grateful for the opportunity to author this second edition. The new edition contains many new figures, tables, and treatment planning cases as well as a comprehensive review of new classifications. I wrote this book because despite a plethora of study materials and information, there is no comprehensive single source study guide to help students prepare for their examinations. *Periodontal Review Q&A* was specifically written to address this void.

The material in this book is presented in a question and answer format for ease of study. The classic literature is cited as well as more recent and practical literature on topics such as diagnosis, nonsurgical therapy, surgical therapy, regeneration, and implants. Literature evidence for opposing viewpoints is also presented throughout the book. Additionally, each chapter contains clear and relevant tables, illustrations, and pictures. This comprehensive and yet concise approach to periodontics is aimed at preparing the candidate for periodontal examinations and clinical practice.

Periodontal Review Q&A is a useful resource for residents, practicing periodontists preparing for board certification, dental students, and dental hygiene students seeking a broader appreciation and in-depth understanding of periodontics. Topics chosen are those emphasized in periodontal residency graduation examinations as well as the oral examination of the American Board of Periodontology. Readers are urged to study all literature preceding their examination, including literature that may be made available subsequent to this textbook's publication.

Acknowledgments

I would like to acknowledge my mentors, Drs Philip R. Melnick, Thomas N. Sims, Paulo M. Camargo, Thomas Han, Henry H. Takei, and Perry R. Klokkevold, for their advice and guidance. Thank you to all the reviewers, Drs Dennis P. Tarnow, Russell Christensen, Jack G. Caton, Michael P. Rethman, Mary E. Neill, and Sejal R. Thacker.

I would like to thank my loving husband, David; my children, Gabriella and Elliot; and my parents. Without their love and support this book would not have been possible.

I appreciate Quintessence and the editorial staff, especially Zachary Kocanda and Bryn Grisham, whose knowledge and dedicated care to every word and idea made this book what it is.

Evidence-Based Dentistry

1

Background

Q: What is the evidence-based approach?

Evidence-based dentistry is the merging of clinically pertinent scientific evidence to the patient's oral and medical condition and history as well as the dentist's experience (Fig 1-1). The dentist uses the evidence to make sound decisions about diagnosis, prognosis, and treatment. Evidence-based decision making consists of formulating patient-centered questions (Population-Intervention-Comparison-Outcome [PICO]); examining and critically evaluating the evidence; and relating the evidence to practice.¹

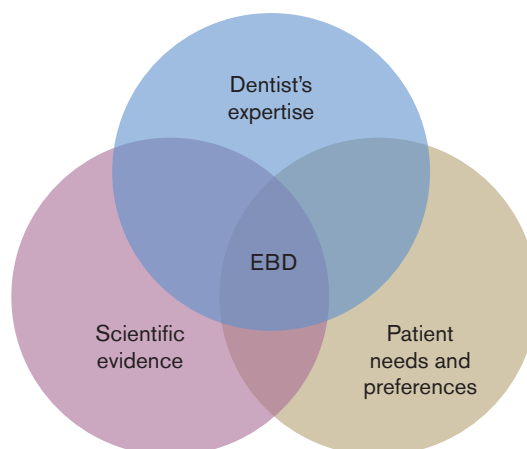


Fig 1-1 Three parts of the decision-making process. (Redrawn from the American Dental Association¹ with permission.)

Q: What is the PICO question?

The *PICO question* is a question that includes a *population* to be examined, the nature of the *intervention* to be inspected, a *comparison* statement, and the type of *outcome* to be evaluated. It should be problem-focused and concise.

Example: In patients with horizontal alveolar ridge deficiencies (population), what is the effect of horizontal bone augmentation procedures (intervention) compared with controls (comparison) on peri-implant health (outcome)?

Q: What is the step-by-step process for making an evidence-based decision in a dental practice?

The steps involved in evidence-based decision making in a dental practice are shown in Fig 1-2.



Fig 1-2 Evidence-based decision making. (Based on data in Chiappelli et al.²)

Studies

Q: What are the different study types (ranked from highest level of evidence to lowest)?

The different types of studies are shown, ranked in order of highest to lowest level of evidence, in Fig 1-3.

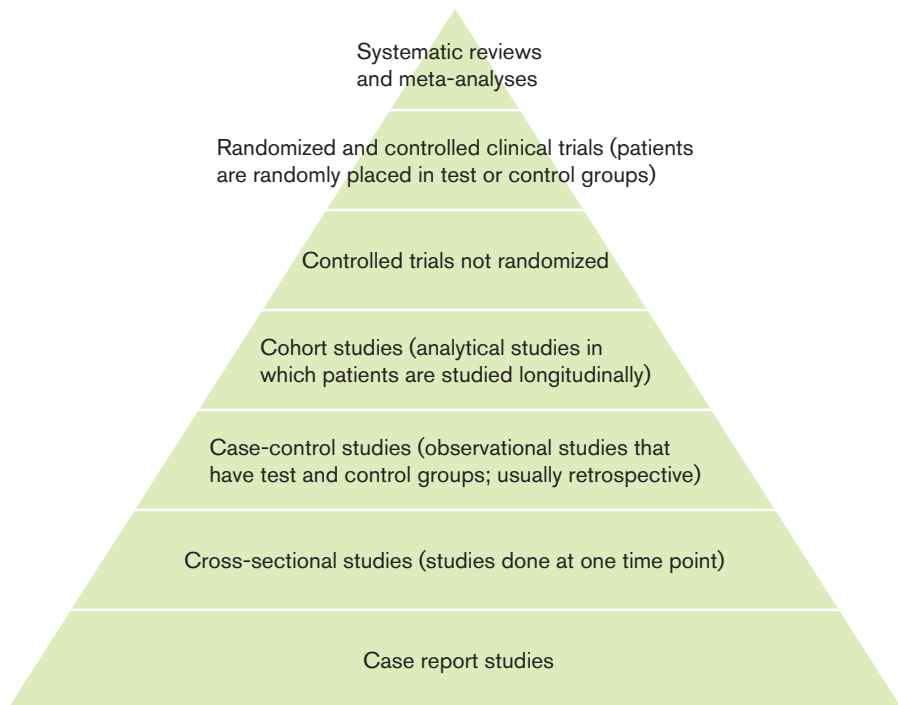


Fig 1-3 Different studies ranked from highest level of evidence to lowest. (Based on Nocini et al.³)

Q: Describe the difference between a cross-sectional study and a longitudinal study.

A cross-sectional study is done at one time point, whereas a longitudinal study ranges over a period, allowing temporal relationships to be investigated.

Q: What is the *P* value?

The *P* value is the probability of obtaining a test statistic at least as extreme as the one observed, assuming that the null hypothesis is true. The smaller the *P* value, the less likely the effect was due to chance. A *P* value less than or equal to .05 usually indicates statistical significance.

Q: What is the difference between sensitivity and specificity?

Sensitivity is the ability of a test to correctly identify diseased individuals.

Specificity is the ability of a test to correctly identify a healthy individual.

For instance, the diagnostic sensitivity of a clinical parameter (suppuration, gingival plaque) in predicting disease was expressed as the proportion of sites showing attachment loss that also exhibited the given parameter. Diagnostic specificity was expressed as the proportion of sites not exhibiting the clinical parameter and not showing attachment loss.⁴

Q: What is the difference between internal and external validity?

The difference between internal and external validity is shown in Fig 1-4.

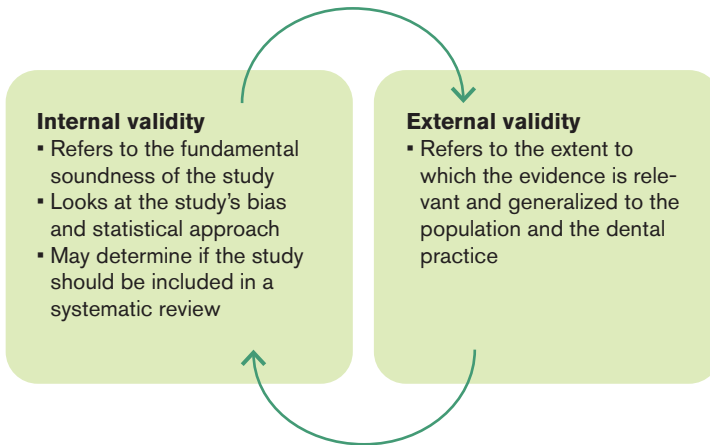


Fig 1-4 Internal and external validity.

Q: Why practice evidence-based dentistry?⁵

1. There are thousands of articles published monthly in dental magazines. It would take hundreds of hours to read the dental literature. Using evidence-based review databases eases the necessary time spent evaluating dental literature.
2. Practicing evidence-based dentistry keeps dentists current on recent evidence and practice standards.
3. A thorough and analytical literature review should be carried out before proceeding in clinical research.

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Periodontal Anatomy

2

Anatomy

Q: Identify the anatomical structures of the periodontium shown below.

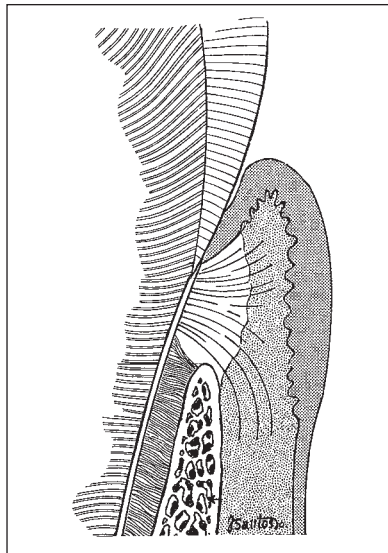


Fig 2-1a Illustration of the periodontium. (Reprinted from Fan and Berry¹ with permission.)

The answers are shown on the next page in Fig 2-1b.

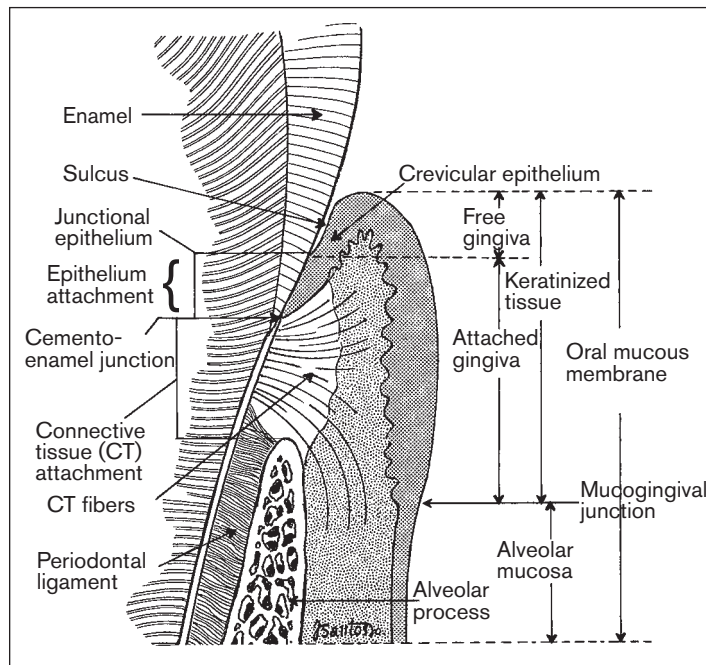


Fig 2-1b Labeled anatomy of the periodontium. (Reprinted from Fan and Berry¹ with permission.)

Q: Where does the vascular supply of the periodontium originate?

The external carotid artery and its main branches, which include the lingual, facial, and maxillary arteries, are the vascular supply for the periodontium. Locally, the blood supply comes from the suprapariosteal vessels and vessels from the periodontal ligament (PDL) and bone.²

Q: What is the main innervation for the periodontium?

The trigeminal nerve and its branches provide the main innervation for the periodontium.

Definitions

Q: What is attached gingiva?

The *attached gingiva* is the area from the base of the sulcus to the mucogingival junction. It prevents the free gingiva from being separated from the tooth. Its height is determined by subtracting the sulcus probing depth from the total width of the keratinized tissue. It consists of thick lamina propria and deep rete pegs. Goasind et al⁹ reported that the attached gingival thickness is 1.25 ± 0.42 mm.

Q: What is keratinized attached gingiva?

The *keratinized attached gingiva* is that found from the gingival margin to the mucogingival junction.

Q: What is alveolar mucosa?

Alveolar mucosa is the covering of the alveolar process that is nonkeratinized, unstippled, and movable. It extends from the mucogingival junction to the floor of the mouth and vestibular epithelium.

Q: What is clinical attachment loss (CAL)?

If the marginal gingiva is below the cementoenamel junction (CEJ):

$$\text{CAL} = \text{pocket depth} + [\text{CEJ to marginal gingiva}]$$

If the marginal gingiva is above the CEJ:

$$\text{CAL} = [\text{marginal gingiva to CEJ}] - [\text{marginal gingiva to bottom of pocket}]$$

Q: What is Ante's law?

Ante's law states that the root surface area of the abutment teeth must be equal to or greater than that of teeth being replaced with pontics. This helps determine the number of abutments needed for a fixed partial denture.

Gingival Epithelium

Q: What are the characteristics of healthy gingiva?

Healthy gingiva is coral pink, firm, follows the CEJ of the teeth, and may be stippled. The color of the gingiva is associated with the pigmentation of the patient. In dark-haired individuals, the gingiva can be darker than that in blond patients.

Q: What are the five types of gingival fibers?

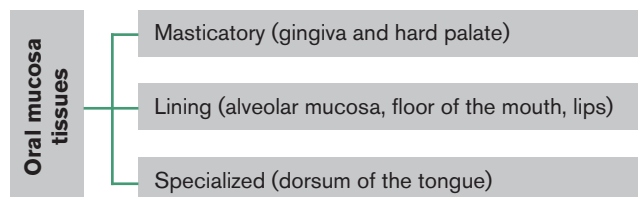
There are five types of gingival fibers:

1. Dentogingival group: There are three types of fibers within this group.
 - Fibers extending coronally toward the gingival crest
 - Fibers extending laterally to the facial gingival surface
 - Fibers extending horizontally beyond the alveolar crest height and then apically along the alveolar bone cortex
2. Alveologingival group: Fibers in this group run coronally into the lamina propria from the periosteum at the alveolar crest.
3. Dentoperiosteal fibers: These fibers insert into the periosteum of the alveolar crest and fan out to the adjacent cementum.
4. Circular group: These are the only fibers that are confined to the gingiva and do not attach to the teeth.
5. Transseptal group: These fibers bridge the interproximal tissue between adjacent teeth and insert into the cementum.

Q: What is the composition of the oral mucosa (the tissue lining the oral cavity)?

The oral mucosa is composed of masticatory, lining, and specialized tissues (Fig 2-2).

Fig 2-2 Composition of the oral mucosa. (Based on Avery.⁴)



Q: What is the composition of the gingival epithelium?

The gingival epithelium consists of oral (masticatory), oral sulcular, and junctional epithelia (average width < 1 mm) (Fig 2-3).

Oral (masticatory) epithelium	Oral sulcular epithelium	Junctional epithelium
▪ Orthokeratinized, stratified squamous ▪ Surface cells lose their nuclei and contain keratin ▪ Made of the following: free gingiva (base of the sulcus to the free gingival margin, ie, the most coronal part of the gingiva), attached gingiva, and palatal tissue	▪ Epithelium that lines the sulcus ▪ No rete pegs in healthy tissue	▪ Attaches to the tooth via a hemidesmosomal layer and a basal lamina ▪ Nonkeratinized and has a fast turnover ▪ Permeable ▪ Most apical part lies at the CEJ in healthy tissue

Fig 2-3 Composition of the gingival epithelium. (Based on Clerehugh et al.⁵)

Q: What are the four layers of cells that comprise the masticatory epithelium?

There are four layers of cells that comprise the masticatory epithelium²:

1. Stratum basale: Cuboidal cells found at the basement membrane; epithelial cell replication takes place in this location. This layer contains melanocytes and Merkel cells.
2. Stratum spinosum: The “spines” are desmosomes allowing intracellular contacts. It is the thickest layer and contains Langerhans cells, which are derived from bone marrow and take part in immune surveillance.
3. Stratum granulosum: Cells in this layer appear flat. Keratinocytes migrating from the underlying stratum spinosum become known as *granular cells* in this layer. These cells contain keratohyalin granules, protein structures that promote hydration and cross-linking of keratin. →

4. Stratum corneum: Outermost layer containing dead cells and consisting of ortho- and parakeratinization. It is composed of compactly packed tonofilaments.

Q: Where are the widest and narrowest zones of gingiva?

The average thickness of the gingiva is 1.25 mm.³ The widest zone of gingiva is in the maxillary anterior region; the narrowest zone is at the facial aspect of the mandibular first premolar.²

Connective Tissue

Q: What is the composition of connective tissue?

Connective tissue (average width slightly greater than 1 mm) is fibrous, consisting of mostly type I collagen, ground substances, and mucopolysaccharides. It also contains white blood cells, blood vessels, lymphatics, and nerves.²

Q: What determines whether epithelium is keratinized or nonkeratinized?

The underlying connective tissue determines whether the epithelium is keratinized.⁶

Q: What is periosteum? What is its function?

The *periosteum* is a highly vascular connective tissue sheath covering the external surface of all bones except areas of articulation and muscle attachment. It consists of an inner cambium layer (contains osteoblasts and osteoprogenitor cells) and an outer fibrous layer.⁷

The periosteum is involved in bone healing and bone regeneration.⁸ It also serves as protection as well as a channel for the blood supply and nutrients for bone tissue.

Periodontal Ligament (PDL)

Q: Where is the average width of the PDL greatest and where is it narrowest?

The width of the PDL is greatest at the apex and narrowest in the middle. Older individuals have thicker fiber bundles in the PDL than younger individuals. The average width of the PDL is 0.2 mm.

Q: What provides the blood supply to the PDL?

Superior and inferior alveolar arteries provide the blood supply to the PDL, which is a vascular tissue.

Q: What are the functions of the PDL?

- Protect vessels and nerves
- Transmit occlusal forces
- Attach the tooth to bone
- Perform formative and remodeling functions

Q: What are the fibers of the PDL?

The fibers of the PDL include the alveolar crest, horizontal, oblique (most numerous), interradicular, and apical fibers.

Q: Describe and define ankylosis.

Ankylosis is the fusion of the cementum and alveolar bone with obliteration of the PDL. It develops after chronic periapical inflammation, tooth reimplantation, and occlusal trauma.

Alveolar Bone

Q: What is the composition of alveolar bone?

Alveolar bone consists of²:

- Cortical bone
- Cancellous trabeculae (more prevalent in the maxilla)
- Alveolar bone proper (lines the tooth socket)

Q: What are the functions of the alveolar bone?

The alveolar bone has three functions²:

1. Protection
2. Support
3. Calcium metabolism

Cementum

Q: Where are acellular cementum and cellular cementum located?

- Acellular cementum is located on the enamel at the CEJ. It does not contain cementocytes and forms slowly.
- Cellular cementum is located at the apical third of the root. It is more irregular and forms rapidly. With age, there is an increase in width of the cellular cementum.

Q: What percentage of the cementum and enamel overlap?

- 60% of the cementum and enamel overlap.
- 30% of the cementum and enamel form a butt joint.
- 10% of the cementum and enamel are separated by a gap.

Q: What is the difference between extrinsic and intrinsic cementum?

Extrinsic fibers are made of Sharpey fibers from the PDL, whereas intrinsic fibers are cementum fibers produced by cementoblasts (Fig 2-4).

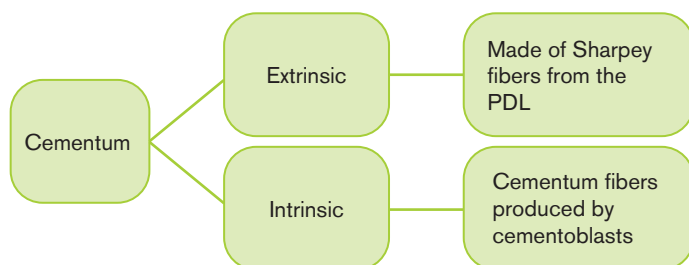


Fig 2-4 Characteristics of extrinsic and intrinsic cementum.

Q: How does the junctional epithelium attach to the cementum?

The junctional epithelium attaches to the cementum via hemidesmosomes and replicates every 5 days.

Temporomandibular Joint (TMJ)

Q: What is the composition of the TMJ disc (meniscus)?

The TMJ disc is composed of dense connective tissue.

Q: Describe the movement of the TMJ.

The meniscus divides the joint into two compartments. The upper compartment has translational movement, and the lower compartment has rotational movement.

Q: What is meniscal derangement with and without reduction?

- With reduction: The disc as well as the posterior band of the meniscus is anteriorly displaced in front of the condyle upon opening. This causes a popping or clicking sound.
- Without reduction: In some patients, the meniscus remains anteriorly displaced at full opening. This is a much more serious condition.

Collagen

Q: Describe the four different types of collagen.

- Type I: Skin, tendon, vascular ligature, organs, bone (main component of the organic part of bone)
- Type II: Cartilage (main component of cartilage)
- Type III: Comprised of reticular fibers, commonly found alongside type I collagen, found mostly in smooth muscle
- Type IV: Forms basis of cell basement membrane

Supracrestal Tissue Attachment (Previously Biologic Width)

Q: What is supracrestal tissue attachment?

The *supracrestal tissue attachment* is defined as the physiologic dimension of the junctional epithelium and connective tissue attachment. It is measured from the most coronal part of the junctional epithelium to the crest of the alveolar bone. In studies on cadavers, Gargiulo⁹ found a connective tissue attachment of 1.07 mm and a junctional epithelium attachment of 0.97 mm. Therefore, the supracrestal tissue attachment is about 2 mm. He also found the sulcus, which is not part of the supracrestal tissue attachment, to be 0.69 mm.

Q: What results from violation of the supracrestal tissue attachment?

If subgingival restorations violate the supracrestal tissue attachment, periodontal bone loss and inflammation may occur. The body will try to make room between the margin of the restoration and the alveolar bone to allow for reestablishment of the supracrestal tissue attachment.

Günay¹⁰ did a study comparing crowns with interproximal margins placed at different distances from the alveolar bone. The three groups were ≤ 1 mm (I), 1 to 2 mm (II), and ≥ 2 mm (III) between crown margin and alveolar crest. It was observed that probing depth and papillary bleeding index was greater for group I (which had an encroachment of its crown margins within the supracrestal tissue attachment).

Miscellaneous

Q: What is the most common area of recurrent pockets?

The mesial aspect of the maxillary first premolars and first molars are the most common areas of recurrent pockets.

Q: What is the relationship between tooth support and root morphology?¹¹

1. Root curvatures and concavities increase periodontal support because they increase the surface area and allow for multidirectional fiber orientation, which makes the tooth more stable.
2. Multirrooted teeth have increased support and resistance to applied forces (the more coronal the furcation, the more stability).
3. Divergent roots increase stability and allow for more bone support.
4. Conical roots have less attachment area and are not as stable.
5. Enamel pearls can weaken periodontal attachment.
6. Root fractures can lead to periodontal destruction.

Many other factors can influence tooth stability, such as inflammation, occlusion, and the density and structure of bone.

Q: Which muscles elevate and depress the mandible?

- Elevate: Temporalis, medial pterygoid, and masseter
- Depress: Lateral pterygoid, digastric, and mylohyoid

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Furcations

3

Background

Q: Define furcation lesion.

A *furcation lesion* has been defined as “the pathologic resorption of bone in the anatomic area of a multi-rooted tooth where the roots diverge.”¹

Q: What is the prevalence of furcation involvement?

In a study on periodontal patients, Svärdröm and Wennström² found a greater prevalence of furcation involvement in maxillary molars than in mandibular molars. The narrowest furcation entrance was found on the buccal aspect of maxillary and mandibular molars, and the highest frequency of involvement was the distal aspect of the maxillary first molar. The mesial aspect of the second molar had the least frequency of furcation involvement.

Q: What is a furcation fornix?

The *furcation fornix* is the roof of the furcation.

Q: Is the Nabers probe a valid and efficient tool for detecting furcation invasion?

Eickholz and Kim³ found that the Nabers probe, marked in 3-mm increments, is a valid method of diagnosing furcation lesions.

Classification

Q: Describe the Hamp classification.

The classification by Hamp et al⁴ involves a horizontal measurement:

- F0: No furcation involvement.
- F1: The probe can penetrate the furcation less than 3 mm.
- F2: The furcation can be probed greater than 3 mm, but it is not a through and through furcation involvement.
- F3: Through and through furcation involvement.

Q: Describe the Glickman classification.

The Glickman⁵ classification is presented in Fig 3-1.

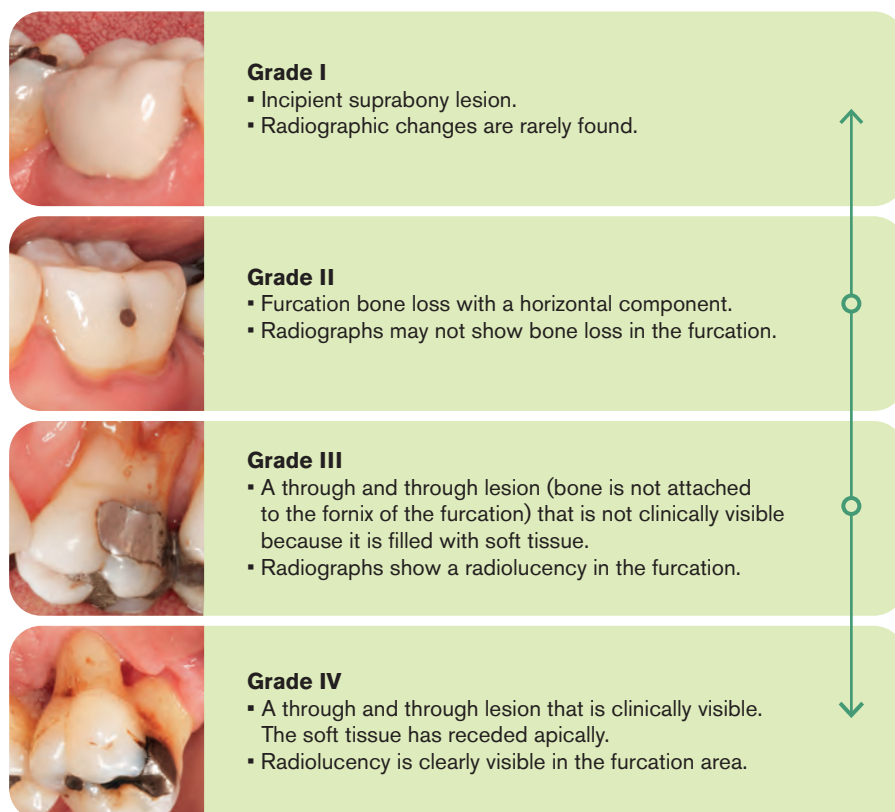


Fig 3-1 Glickman classification.

Q: Describe the Tarnow classification.

The classification by Tarnow and Fletcher⁶ is a subclassification of the Glickman furcation classification that measures the vertical probing depth from the roof of the furca:

- A: 0 to 3 mm
- B: 4 to 6 mm
- C: Greater than 7 mm

Treatment Options

Q: What are the treatment options for furcation defects?

- Nonsurgical debridement
- Surgical debridement
- Surgical exposure of the furcation
- Regeneration (guided tissue regeneration [GTR] and enamel matrix derivative [EMD])
- Extraction
- Root resection
- Tunnel preparation

Q: Is open/closed flap scaling and root planing effective in furcation lesions?

A review by Cobb⁷ demonstrated a less favorable response to scaling and root planing by molars with furcation involvement compared with those without furcation lesions and single-rooted teeth. He surmised that this was related to the inability to remove all pathogenic microbial flora due to the furcal anatomy restricting access for mechanical therapy.

Bower⁸ found that 81% of the time the furcation entrance is 1 mm or less. The study also found that 58% of the time the furcation entrance is 0.7 mm or less. The blade width of commonly used periodontal curettes is 0.75 mm. The ultrasonic (smaller) tip would fit better than the tip of a Gracey curette in a grade II or III furcation.

Wylam et al⁹ found no significant difference between open and closed flap root planing. The study further concluded that root planing is inefficient in the debridement of furcation lesions and does not allow for periodontal regeneration.

Q: What are guidelines for root resection?¹⁰

Situations favorable for root resection:

- The candidate tooth is of critical importance to the overall dental treatment plan (abutments, important to a prosthesis).
- The tooth has enough attachment present to function.
- No other cost-effective therapy is available.
- Patient has good oral hygiene.

How to determine which root to remove:

- Remove the root that eliminates the furcation.
- Remove the root that has the greatest amount of bone and attachment loss.
- Remove the root that eliminates periodontal bone loss on adjacent teeth.
- Remove the root with the most anatomical problems.
- Remove the root that would complicate future treatment the least.
- The most common root resected is the distobuccal root of the maxillary first molar.

See chapter 10 for more information—eg, indications, contraindications—on root resections.

Q: Would you have endodontic therapy done before a root resection and why?

Endodontic therapy should be done prior to root resection for the following reasons:

- Little/no pain.
- It would be prudent to do the root canal therapy so that if a perforation, fracture, or other adverse outcome occurs, the patient does not have to have the resection done.¹⁰

Q: Describe the technique for root resection.¹⁰

1. Facial and lingual flaps are elevated.
2. Debridement and exposure of the furcation of the root to be resected.
3. Removal of a small amount of buccal or palatal bone to allow access for elevation and root removal.
4. An oblique cut is made with the high-speed handpiece using a carbide bur (surgical length) through the tooth.
5. The root is elevated from the socket, and if necessary odontoplasty and/or periodontal surgery are done on adjacent teeth. →

6. Sutures should be placed.
7. Occlusion may have to be adjusted because of removal of the root.

Q: Can a grade I, II, III, IV furcation be treated successfully? Which treatment is the most effective?

According to the 2015 consensus report from the American Academy of Periodontology¹¹ (Table 3-1):

Table 3-1 Most effective treatments	
Hamp classification	Most effective treatment
F1	Regenerative therapy may be helpful, although generally F1 furcation defects may be treated successfully with non-regenerative therapies.
F2	Regenerative periodontal therapy should be considered before resective therapy or extraction. A combined therapeutic approach (membrane, bone graft with or without biologics) appears more beneficial over monotherapeutic algorithms.
F3	Proof of histologic periodontal regeneration in mandibular F3 defects is limited to one case report. Favorable outcomes after regenerative therapy for maxillary F3 furcation defects are limited to clinical case reports.

According to a systematic review by Huynh-Ba et al,¹² treatment for the maintenance of teeth with furcation defects (ie, open flap debridement [OFD], tunneling, root amputation, hemisection, and regeneration) are associated with high survival rate ranges, with GTR being the highest one with a survival rate range of 83.3% to 100% after 5 to 12 years. These survival rates are pretty close to implant survival rates in periodontal patients for the same time period (see implant chapter).

Huynh-Ba et al¹² also found the most recurrent complications after resective treatment were not related to periodontal disease but were related to endodontic failures and vertical root fractures. It is important to note that full furcation closure was not predictably attained following GTR or the application of EMD in maxillary and mandibular molars. →

Kinaia et al¹³ did a meta-analysis and discovered, “guided tissue regeneration with the use of resorbable membranes was superior to non-resorbable membranes in vertical bone fill. Both types of membranes were more effective than open flap debridement in reducing vertical probing depths and gaining vertical attachment levels and in gaining vertical and horizontal bone.”

Chen et al¹⁴ did a meta-analysis and found that the GTR and GTR with osseous grafting (OG) groups obtained greater vertical/horizontal bone fill, furcation closure rate, and vertical/horizontal attachment level gain than the OFD group in mandibular molars. The GTR group obtained greater vertical attachment level gain and vertical/horizontal bone fill than the OFD group in maxillary molars. The GTR with OG group achieved better clinical outcomes than the GTR group did in all the studies comparing outcomes in mandibular molar.

Similarly, Murphy and Gunsolley¹⁵ performed a systematic review and discovered that GTR is more effective than OFD in the treatment of furcation defects. The study also found that in furcation lesions vertical probing depth, vertical probing attachment levels, and horizontal open probing attachment levels—but no intrabony outcomes—were improved by augmentation with a particulate graft in conjunction with the GTR barrier.

Evans et al¹⁶ reviewed 50 papers that studied 1,016 grade II furcations. The teeth had received various treatments such as bone grafts, OFD, and GTR. The study found complete closure of the furcation in 20% of the grade II furcations and an improvement of a grade II to a grade I furcation in 33% of the cases. In general, there was a 50% improvement for grade II furcations. The most effective treatment was a bone graft with GTR, and the least effective treatment was OFD (2% of the grade II furcations had complete closure).

Bowers et al,¹⁷ who treated his patients with a combination therapy using an expanded polytetrafluoroethylene (ePTFE) membrane and demineralized freeze-dried bone allograft (DFDBA), discovered complete closure of grade II mandibular furcations in 50% of patients. There was a 68% improvement of a grade II to a grade I furcation.

Q: What factors influence the success of treatment?

Bowers et al¹⁷ found poorer results in the treatment of furcation lesions in smokers. Smokers had a 62.5% chance of having a grade II residual defect compared with nonsmokers, who had a 14.3% chance. Furcation fill decreases at an increased horizontal and vertical presurgical probing attachment level (greater than 5 mm). The following were found to reduce the frequency of clinical closure: →

- Increased distance between the roof of the furcation and the crest of bone
- Increased distance between the roof of the furcation and the base of the defect
- The depth of the horizontal defect
- Increased divergence of roots at the crest of bone

Q: Are teeth that undergo the tunnel preparation in the furcation area prone to caries?

Hellden et al¹⁸ studied 156 teeth with advanced periodontal furcation defects that were treated by tunnel preparations. The study found that 25% of the teeth developed caries.

A retrospective study by Feres et al¹⁹ demonstrated that a history of root caries was the only factor with a positive association with caries incidence in tunnels.

Q: Are biologic mediators effective in furcation lesions?

Enamel matrix derivative (EMD)

Because grade II or III furcation defects are noncontained defects, the use of biologic materials has the significant limitation that, because of their fluid or gel-like quality, any space-making effect is prevented, and therefore the regenerative possibility of such tools may be inadequate in furcation lesions.¹

A study by Araújo and Lindhe²⁰ used EMD and a barrier in the test defects and just a barrier in control defects. The test defect was found to have acellular cementum in the apical portion, while in the coronal portion a thick cellular cementum, similar to the cementum found in the control group, was detected. Both the test and control group furcation defects were found to be clinically closed and to contain bone and periodontal ligament tissue that appeared structurally similar to newly formed root cementum.

A randomized clinical trial by Jepsen et al²¹ comparing membrane placement and EMD treatment of buccal grade II furcation involvement in mandibular molars found a significantly greater reduction in horizontal furcation depth and a comparatively lower incidence of postoperative swelling/pain following EMD compared to membrane therapy.

Although enamel matrix protein therapy has exhibited clinical improvements in the treatment of buccal grade II furcation defects in mandibular molars, complete closure of the furcation lesion is attained only in a minority of cases.¹



Recombinant human osteogenic protein-1 (OP-1)

Giannobile et al²² found distinct stimulation of osteogenesis, regenerative cementum, and new attachment formation in sites treated with three different concentrations of OP-1.

Platelet-rich plasma (PRP)

Pradeep et al²³ reported that although the grade II status of the studied furcation defects was unchanged, sites treated with PRP showed a statistically significant difference in all clinical and radiographic parameters compared with those receiving OFD.

Anatomical Factors

Q: What anatomical factors are associated with furcation lesions?

The anatomical factors associated with furcation lesions are shown in Fig 3-2.

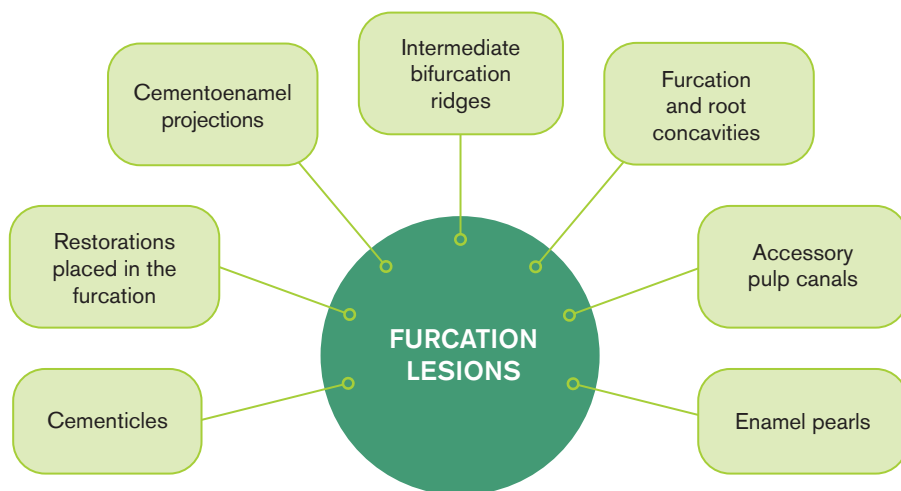


Fig 3-2 Anatomical factors associated with furcation lesions.

Q: What percentage of molars have accessory canals in the furcation?

Gutmann²⁴ studied 102 molars and found that 28.4% (29.4% of mandibular molars and 27.4% of maxillary molars) have accessory canals in the furcation.

Q: What is the classification for cervical enamel projections?

Masters and Hoskins²⁵ classified cervical enamel projections as follows:

- Grade 1: Extends from the CEJ to the furcation
- Grade 2: Approaches the furcation
- Grade 3: Extends into the furcation horizontally

Q: What is the frequency of cervical enamel projections?

Swan and Hurt²⁶ discovered that mandibular second molars had the highest incidence of cervical enamel projections (51.0%), while the lowest incidence was found in the maxillary first molars (13.6%). The most frequently encountered enamel projections in the study were grade 1, and the buccal surfaces were the most common location of cervical enamel projections.

Roussa²⁷ observed 60 teeth and discovered cervical enamel projections in 30% of the teeth examined.

Q: Is there a relationship between cervical enamel projections and furcations?

It is believed that the existence of cervical enamel projections inhibits connective tissue attachment and makes management of the furcation difficult.

Swan and Hurt²⁶ showed a positive correlation between grade 2 and grade 3 cervical enamel projections and periodontally involved furcations. However, the study did not find a relationship between grade 1 projections and furcation involvements.

A review by Lima and Hebling²⁸ found an association between cervical enamel projection, inflammatory periodontal disease, and molar furcation involvement.

Q: Describe how root concavities can affect prognosis of a tooth. Where are the concavities located on the first molars?

It is much more difficult to perform scaling and root planing on teeth with concavities, and as a result they can have a deleterious effect on the prognosis of a tooth. It is also challenging for the patient to maintain oral hygiene.

The root concavities for the first molars, as described by Bower,²⁹ are presented in Fig 3-3.

They are present in all two-rooted maxillary premolars.

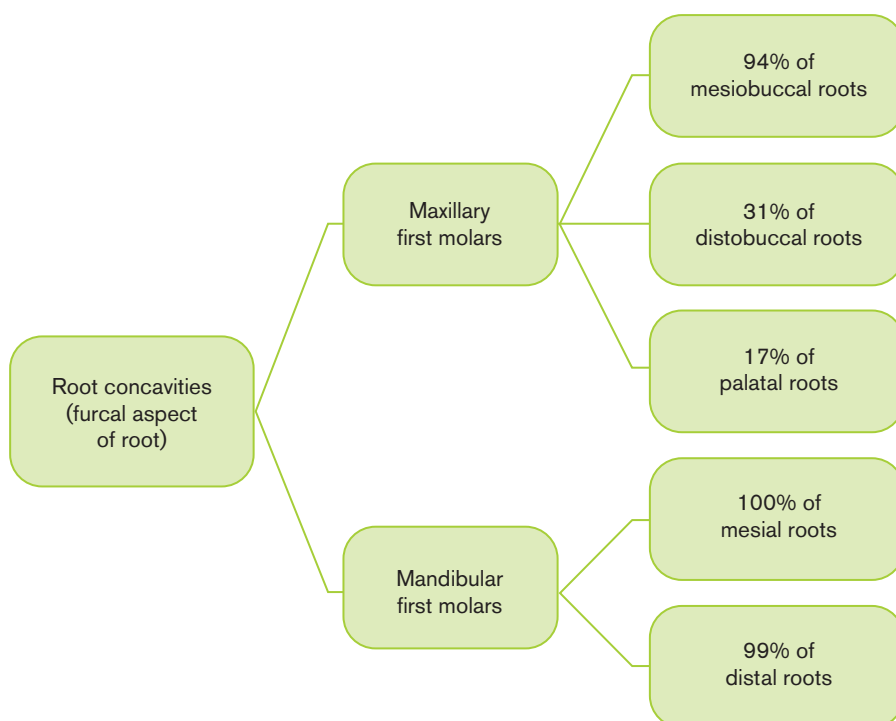


Fig 3-3 Anatomy of root concavities.

Q: How do different furcations compare in regard to entrance width?

The differences in furcation entrance width are as follows²⁹:

- Mandibular molars: Buccal furcation entrance width is greater than lingual furcation entrance width. Further, the buccal furcation entrance in the mandibular first molar is positioned close to the cemento-enamel junction (CEJ), whereas it is more apically located in the mandibular second molars.¹



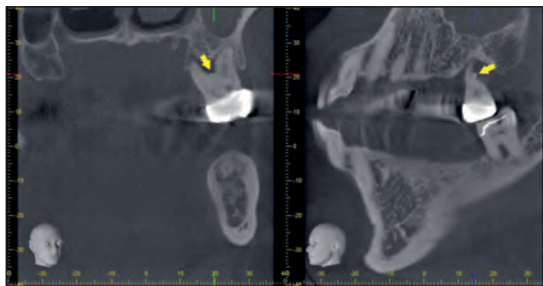
- Maxillary molars: Mesial entrance width is greater than distal entrance width, which is greater than buccal entrance width. The mesial–palatal furcation entrance of the maxillary first molar is located close to the palatal third of the tooth, whereas the distal–palatal furcation is in the middle portion of the tooth.¹

**Q: Where are enamel pearls (see Fig 3-4) found most often?
How can they contribute to furcation lesions?**

According to Moskow and Canut,³⁰ enamel pearls (spherical in shape) have a tendency to originate in the furcation areas of molars, especially the maxillary second and third molars. The study found an incidence of 1.1% to 9.7%.

As with cervical enamel projections, the existence of enamel pearls inhibits connective tissue attachment and makes management of the furcation difficult.

Fig 3-4 Cone beam image of enamel pearl in the furcation of the mandibular left second molar.



Q: How much attachment loss occurs for a furcation entrance to be observed in a maxillary molar?

According to Gher and Dunlap,³¹ the mean distances from the CEJ to the mesial, facial, and distal furcation entrances are 3.6, 4.2, and 4.8 mm, respectively. Attachment loss of 6 mm is associated with a grade III furcation.

Q: How much surface area does each of the maxillary roots comprise?

According to Hermann et al,³² the roots comprise the following surface areas:

- Mesiobuccal: 36%
- Palatal: 35%
- Distobuccal: 28%

Q: What is an intermediate bifurcation ridge? How can it contribute to a furcation lesion?

Everett et al³³ first described the ridges. He studied 328 mandibular teeth and found that 73% had an intermediate bifurcation ridge, which is cementum extending from the mesial to the distal of a furcation opening on a mandibular molar. Its existence can hamper effective plaque control by the patient and the dentist.

Q: Describe cementicles and their contribution to furcation lesions.

Cementicles are calcified masses adherent to or separated from the tooth's root. Similar to intermediate bifurcation ridges, they compromise effective scaling and root planing by the dentist and oral hygiene by the patient.

Restorations

Q: Do restored molars have a higher percentage of furcation lesions than teeth without restorations?

Wang et al³⁴ demonstrated a higher percentage of furcation involvement but no greater mobility in molars with a crown or proximal restoration when compared with nonrestored molars. Restored molars also showed a greater mean probing periodontal attachment loss than nonrestored molars, but the difference was only marginally significant.

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Epidemiology and Etiology

4

Risk Factors and Risk Indicators

Q: What is the difference between a risk factor and a risk indicator?

A *risk factor* is a characteristic that places an individual at increased risk of contracting a disease. A *risk indicator* is a probable or putative risk factor that has been identified in cross-sectional correlation studies but not confirmed through longitudinal studies.

Q: What are the risk factors for gingivitis and periodontitis?

Poor oral hygiene

- Listgarten¹ suggested increased plaque mass or reduced host defense may precipitate episodes of periodontal destruction.
- Löe et al² observed 480 tea plantation workers that were not exposed to preventive care for 15 years. They found that 8% of the population had rapid progression to periodontal disease, 11% had no progression, and 81% had moderate progression.

Dental plaque biofilm retention factors

- Certain tooth anatomical factors that facilitate plaque accumulation at and apical to the gingival margin (eg, enamel pearl).
- Subgingival restoration margins may be detrimental to gingival and periodontal health.³ Another study found that when plaque control →

and periodontal maintenance measures are described, and the patient complies, subgingival margins do not act as a plaque retentive factor that causes gingivitis.⁴

- A direct or indirect restoration with overhanging margins can be associated with localized gingivitis, increase in pocket depth, and interproximal bone loss.⁴
- Tooth position (eg, crossbite, malalignment, and crowding).⁴

Oral dryness

- Reduced cleansing, which can lead to increased dental plaque and enhanced gingival inflammation.

Smoking

- Strongest modifiable risk factor and predictor of future disease.
- According to Grossi,⁵ smokers are 2.7 times more likely to have periodontal disease than nonsmokers.
- Smokers have 18 times more periodontal pathogens compared with nonsmokers.

Diabetes

- Three times more bone loss and attachment loss has been found in patients with diabetes.
- After treatment of periodontal disease, there is a 10% drop in sugar levels.⁶

Genetics

- In a study by Kornman et al,⁷ 86.0% of the severe periodontitis patients were either smokers or had the interleukin-1 genotype.
- According to Michalowicz et al,⁸ 50% of enhanced risk for periodontitis can be accounted for by genetics alone.

Q: What are the risk indicators for periodontitis?

Following are the risk indicators for periodontitis:

- Age: Increased age is associated with increased extent and severity of chronic periodontitis.
- Sex: Male patients have poorer hygiene than female patients.
- Socioeconomic status: Those with lower socioeconomic status have less education and limited access to care. →

- **Race:** There is a greater incidence of periodontitis in black and Hispanic populations. Blacks were found to be at much greater risk of aggressive periodontitis than whites.⁹
- **Obesity:** Adipose tissue can produce cytokines, which can lead to a hyperinflamed state. Gorman et al¹⁰ found that overall obesity and central adiposity are associated with an increased hazard of periodontal disease progression in men.
- **Alcohol:** Shepherd¹¹ found evidence suggesting that alcohol consumption is a risk indicator for periodontitis. Gay et al¹² found alcohol consumption was associated with an increase in the likelihood of having periodontitis, especially severe periodontitis.
- **Stress:** Genco et al¹³ found that financial stress and the coping mechanism of the patient were related to periodontal disease. They found that the effects of stress on periodontal disease can be moderated by adequate coping behaviors.
- **Contraceptives:** There is a possible association between injectable progesterone contraceptives and poor periodontal health, as indicated by a study that found an increased likelihood of indicators of poor periodontal health, including gingivitis and periodontitis, in women who had taken or were currently taking depot medroxyprogesterone acetate (DMPA) injectable contraceptives.¹⁴
- **Recreational cannabis:** Researchers have discovered that recurrent recreational cannabis use—which includes marijuana, hashish, and hash oil—may be linked with elevated risk of periodontal disease. When compared to study participants who used cannabis less regularly, those who had taken it at least once a month for a year showed increased indicators of mild, moderate, and severe periodontal disease. The National Health and Nutrition Examination Survey 2011 to 2012¹⁵ found frequent recreational cannabis is associated with deeper pocket depths, more clinical attachment loss, and higher odds of having severe periodontitis.

Epidemiology Terminology

Q: What is an odds ratio?

An *odds ratio* is the probability that a person with an adverse outcome was exposed to risk. An odds ratio that is greater than 1 has a positive association.

Q: What is the difference between sensitivity and specificity?

Sensitivity is the proportion of subjects with a disease who test positive, and *specificity* is the proportion of subjects without the disease who test negative. (See chapter 1.)

Q: What is the difference between prevalence and incidence?

Prevalence is defined as the total number of cases (or cases of the risk factor) in the population at a given time, divided by the number of individuals in the population. *Incidence* is a measure of the risk of developing a new condition within a specified period of time.

Prevalence is a good measure for generalized chronic periodontitis since it develops slowly over time. Incidence is a better tool for toothache since it usually has a rapid onset; however, a specified period of time measurement might miss a lot of cases.

Indices

Q: What is gingival crevicular fluid?

The volume of gingival crevicular fluid (GCF) has been adopted in clinical trials to measure the severity of gingival inflammation.¹⁶ There is a strong relationship between GCF volume and other clinical parameters of gingival inflammation. Measuring GCF is a very reliable quantitative method to assess plaque-induced inflammation in a research setting; however, in clinical practice, measurement of GCF is challenging, time consuming, and costly.

Q: What is the Gingival Index?

The Gingival Index (GI; Löe¹⁷) incorporates bleeding on probing (BOP) and color change:

- 0: Normal gingiva
- 1: Mild inflammation, slight color change, slight edema, and no BOP
- 2: Moderate inflammation, redness, edema, glazing, and BOP
- 3: Severe inflammation, marked redness, edema, ulceration, and spontaneous bleeding →

BOP sites (GI = 2) have a higher chance of attachment loss compared to non-bleeding sites (GI = 0 or 1).¹⁸

The average GI in a representative American adult population that approximates a recent US Census is 1.055, with 93.9% of subjects having a GI $\geq$ 0.50 and 55.7% $\geq$ 1.00.¹⁹

Q: How is BOP assessed?

A BOP score is assessed as the proportion of bleeding sites (no/yes evaluation) when a standardized probe with a 25-g force is placed at the bottom of the sulcus at six sites (distobuccal, buccal, mesiobuccal, distolingual, lingual, and mesiolingual) on all present teeth.^{16,20}

Q: What factors can increase BOP? Decrease BOP?

See Fig 4-1 below.

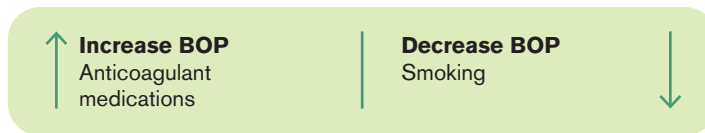


Fig 4-1 Factors that increase and decrease BOP.

Q: What is the modified Gingival Index?

Following is the modified Gingival Index:

- 0: Absence of inflammation
- 1: Mild inflammation, slight color change, and slight edema but not involving the entire marginal or papillary unit and no BOP
- 2: Mild inflammation, slight color change, and slight edema involving the entire marginal or papillary unit and no BOP
- 3: Moderate inflammation, redness, edema, glazing, BOP, and hypertrophy of the entire marginal unit
- 4: Severe inflammation: Marked redness and edema, ulceration with spontaneous bleeding

Q: How early can BOP occur?

BOP can occur as early as 2 days after gingivitis begins.

Q: Describe the Periodontal Index.

In the Periodontal Index (Russell²¹), the supporting tissues around the tooth are scored. Periodontal probing is not recommended.

- 0: Negative
- 1: Mild inflammation
- 2: Gingivitis
- 6: Gingivitis with pocket formation
- 8: Advanced destruction with loss of masticatory function

Q: What is the Periodontal Disease Index? What teeth are used in the index?

The teeth used in the Periodontal Disease Index (Ramfjord²²) are the maxillary right first molar, left central incisor, and first premolar as well as the mandibular left first molar, right central incisor, and first premolar. The cementoenamel junction is used as the fixed landmark to measure periodontal attachment loss.

- G0: Absence of disease
- G1: Mild to moderate inflammation not extending around the tooth
- G2: Mild to moderate inflammation extending around the tooth
- G3: Severe gingivitis and redness

Q: What is the Plaque Index?

The Plaque Index (Löe¹⁷) is based on the thickness of soft deposits found around a tooth by running a probe along the gingival crevice.

- 0: No plaque
- 1: Film of plaque
- 2: Moderate accumulation of plaque within the gingival pocket or on the tooth and gingival margin that can be seen with the naked eye
- 3: Abundance of plaque within the gingival pocket or on the tooth and gingival margin

Prevalence of Gingivitis, Periodontitis, and Gingival Recession

Q: What is the prevalence of gingivitis?

Albandar and Kingman²³ found a prevalence of 32.3% (limited, 21.8%; extensive, 10.5%).

Mild localized clinical inflammation is found in 95% of the population.²⁴

Q: What percentage of tooth extractions are caused by periodontitis?

According to Brown et al,²⁵ less than 20% of all missing teeth were recorded as missing as a result of periodontal disease.

Overall, in the general population, including people with and without periodontitis, mean annual attachment loss was 0.1 mm per year, and mean annual tooth loss was 0.2 teeth per year. Attachment loss per year for periodontitis patients is 0.6 mm.²⁶

Q: What percentage of the population has aggressive, chronic, and severe periodontitis?

- Severe periodontitis is the sixth most prevalent disease of mankind, and it affects 11% of the world population with the prevalence increasing with age. The peak global incidence is at age 38.²⁷
- Localized aggressive periodontitis: 0.53% (greater prevalence in blacks⁹; many studies have proposed that there may be a defect in neutrophil function)
- Generalized aggressive periodontitis: 0.13%

The 2009–2014 National Health and Nutrition Examination Survey²⁸ found the following data on the prevalence of periodontitis among adults in the United States:

- 42.2% of dentate adults 30 years or older had periodontitis: 7.8% with severe periodontitis and 34.4% with nonsevere (ie, mild to moderate) periodontitis.
- 3.3% of all periodontally probed sites or 9.1% of all teeth had probing depth of 4 mm or greater. →

- 19% of sites or 37.1% of teeth had clinical attachment loss of 3 mm or greater.
- Severe periodontitis was most prevalent among Mexican Americans, non-Hispanic blacks, smokers, and adults 65 years or older.
- Periodontitis significantly co-occurred with diabetes and increasing number of missing teeth but not obesity.

Q: What percentage of adults have gingival recession?

Cross-sectional epidemiologic studies have shown that 88% of people 65 years of age and older and 50% of people 18 to 64 years of age have one or more sites with recession.²⁹

Association of Periodontitis with Other Conditions

Q: Is there a relationship between periodontitis and preterm births and low birth weight?

A *preterm birth* is defined as one that occurs before 37 weeks gestation; low birth weight is less than 2,500 grams. According to Offenbacher et al,³⁰ the incidence of preterm birth was 11.2% among women with no periodontal disease and 28.6% in women with moderate to severe periodontitis. Michalowicz et al³¹ found treatment of periodontitis in pregnant women was safe and improved periodontitis, but it did not alter rates of preterm birth. Furthermore, nonsurgical periodontal therapy, ie, scaling and root planing, does not improve birth outcomes in pregnant women with periodontitis.³²

López et al found: “The results of epidemiological, molecular, microbiological and animal-model studies support a positive association between maternal periodontal disease and preterm birth. However, the results of intervention studies carried out to determine the result of periodontal treatment on reducing the risk of preterm birth are controversial.”³³

Figure 4-2³⁴ presents the current theory of the relationship between periodontitis and preterm birth.

Ide and Papapanou³⁵ found preeclampsia is significantly associated with maternal periodontitis. →